



TEAGUARD AI

Intelligent Decision Support System for Sri Lankan Tea Estate Management

PROGRESS PRESENTATION 2



Agenda

What we'll cover today in our second progress review.

- 01** Introduction
- 02** Problem & Measurable Pain Point
- 03** Research Gap & Objectives
- 04** Solution Overview & Architecture
- 05** Integration & Current Progress
- 06** Evidence & Quality Evaluation
- 07** Commercialization
- 08** Validation
- 09** Gantt Chart

THE TEAM

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Smart Tea Estate Management in Sri Lanka

Ceylon Tea is one of Sri Lanka's most valuable exports and a major source of rural employment. Yet most estates still run on manual record-keeping and experience-based decisions leaving managers without timely information on labor, disease, weather, or quality.

TeaGuard AI unifies Artificial Intelligence, Machine Learning, Deep Learning and IoT into one decision-support platform built for the realities of Sri Lankan tea estates.

4

Integrated intelligent modules

3

Tea-growing regions covered

1

Unified web dashboard

Four Compounding Problems on Every Estate



Labor & Welfare Issues

- Labor shortages reduce productivity.
- Manual worker allocation is time-consuming.
- No real-time worker health monitoring.



Tea Leaf Diseases

- Diseases reduce tea yield and quality.
- Early detection is difficult.
- Delayed treatment increases crop losses.



Weather Uncertainty

- Weather affects tea growth and harvesting.
- Manual planning is unreliable.
- No localized weather recommendations.



Manual Tea Grading

- Depends on human experience.
- Inconsistent grading results.
- Slower quality assessment process.

One Fragmented Field, One Unified Platform



The Research Gap

Prior work treats labor management, disease detection, weather prediction and tea grading as isolated problems each solved by a separate, single-purpose system.

No existing platform links these four functions together, so estate managers still stitch decisions across disconnected tools and manual records.

Our response: integrate all four into a single Decision Support System.

Research Objectives

- 1 Optimize workforce allocation with ML while monitoring worker health via IoT wearables (HR, SpO2).
- 2 Detect and classify tea leaf diseases from field images using YOLOv8s, with treatment guidance.
- 3 Forecast weather and predict disease risk per region using Gradient Boosting and Random Forest models.
- 4 Automate tea grade classification using deep learning to improve consistency and reduce processing time.
- 5 Unify all four modules into one real-time dashboard for estate managers.

TeaGuard AI: Four Modules, One Platform

A web-based platform (React.js frontend, Node.js backend) that brings together labor, disease, weather and quality intelligence for estate managers, supervisors and field officers.



1. Labor Management & Welfare

ML workforce allocation + IoT wearables (HR / SpO2)



2. Tea Leaf Disease Detection

YOLOv8s detection with treatment recommendations



3. Weather & Climate Intelligence

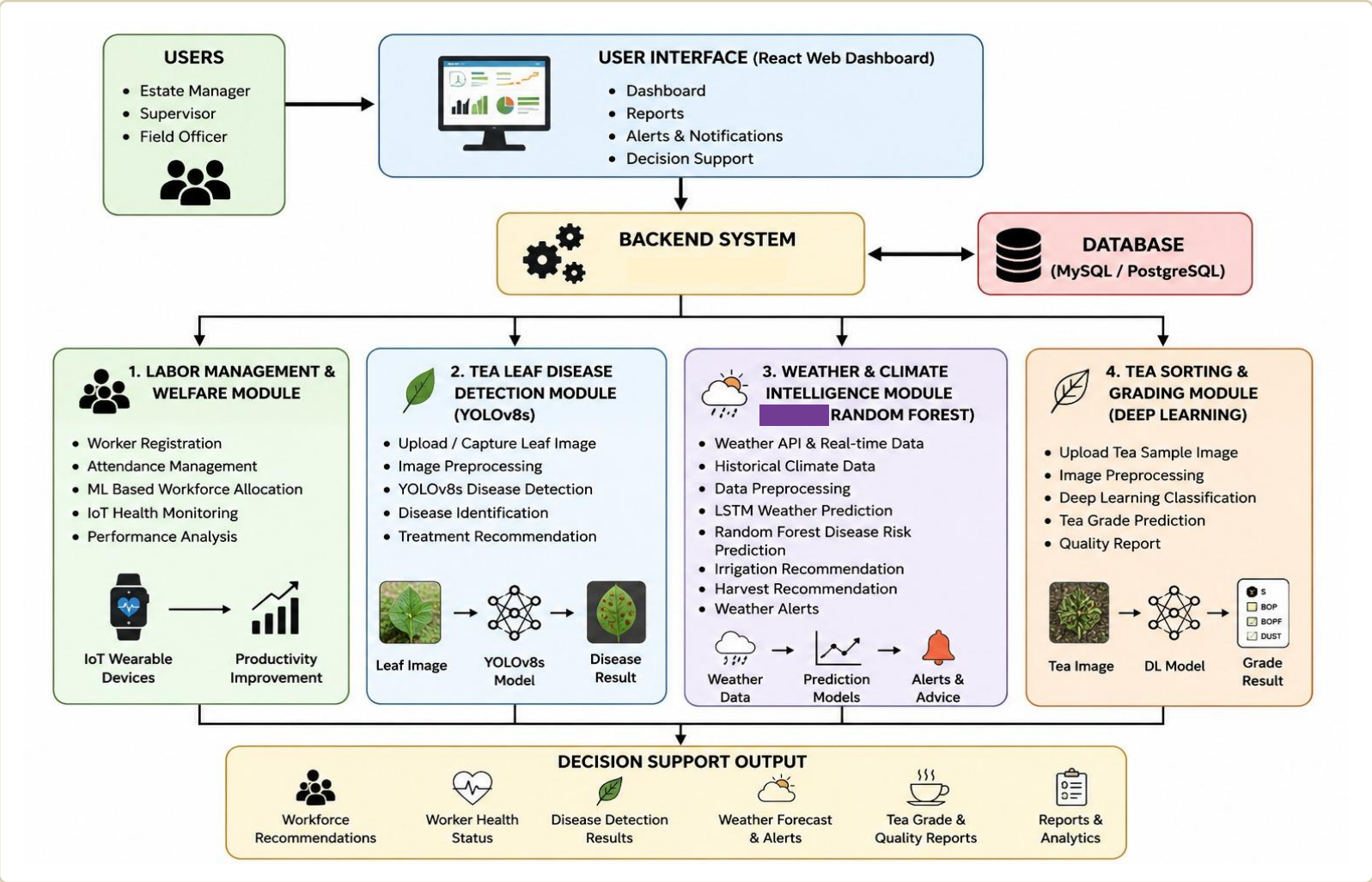
Linear Regression + Random Forest disease-risk model



4. Tea Sorting & Grading Automation

ConvNeXtV2-Tiny deep learning grade classification

Overall System Architecture



01

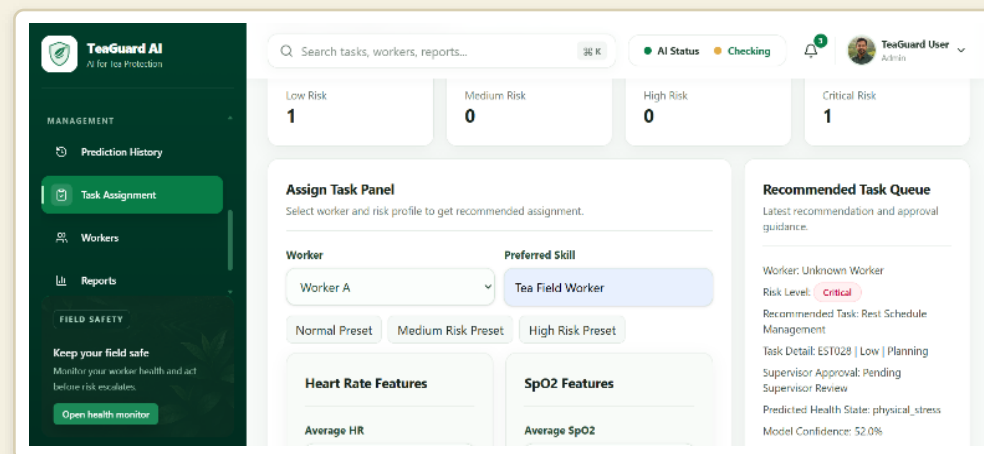


Labor Management & Welfare Optimization

Combines machine-learning workforce allocation with IoT wearable health monitoring, so managers get both better task assignment and real-time visibility into worker wellbeing.

SOLUTION OVERVIEW & ARCHITECTURE

- Worker profiles: experience, skills, availability, past work
- ML-based job assignment for even, skill-matched workloads
- Continuous Heart Rate & SpO2 tracking via wearables
- Automatic alerts when readings cross safe thresholds



Live dashboard preview

82%

Accuracy

LightGBM Classifier

Dataset: PhysioNet Non-EEG Dataset

Relaxed / Emotional Stress / Cognitive Stress / Physical Stress

02

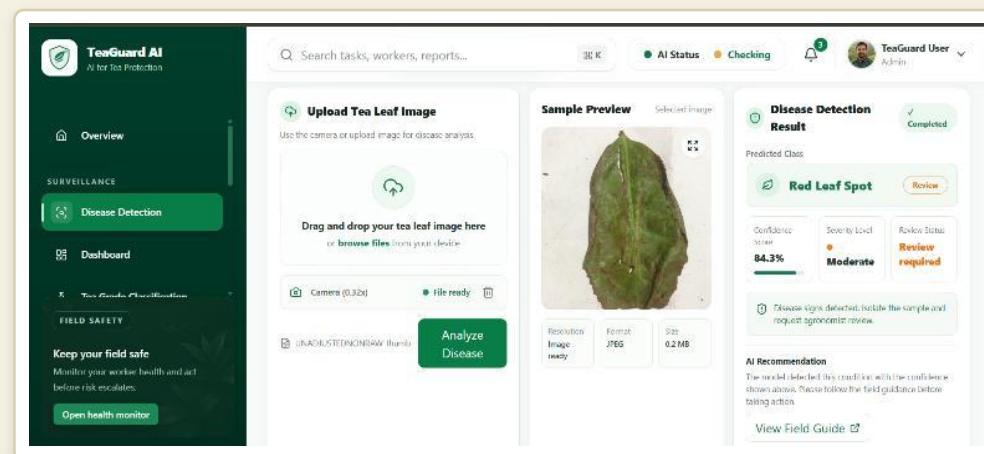


Tea Leaf Disease Detection & Pattern Analysis

Detects tea leaves in an uploaded image, identifies the disease affecting it, and returns a treatment recommendation helping managers respond before an outbreak spreads.

SOLUTION OVERVIEW & ARCHITECTURE

- Leaf presence check before disease analysis begins
- YOLOv8s model trained on a tea leaf disease image dataset
- Disease name + confidence score returned per image
- Built-in treatment advice for the detected disease



Live dashboard preview

94.5%

F1-Score

YOLOv8s (via Hugging Face API)

Dataset: Tea Leaf Disease Image Dataset

95.2% Precision · 93.8% Recall · 94.5% F1-score

03

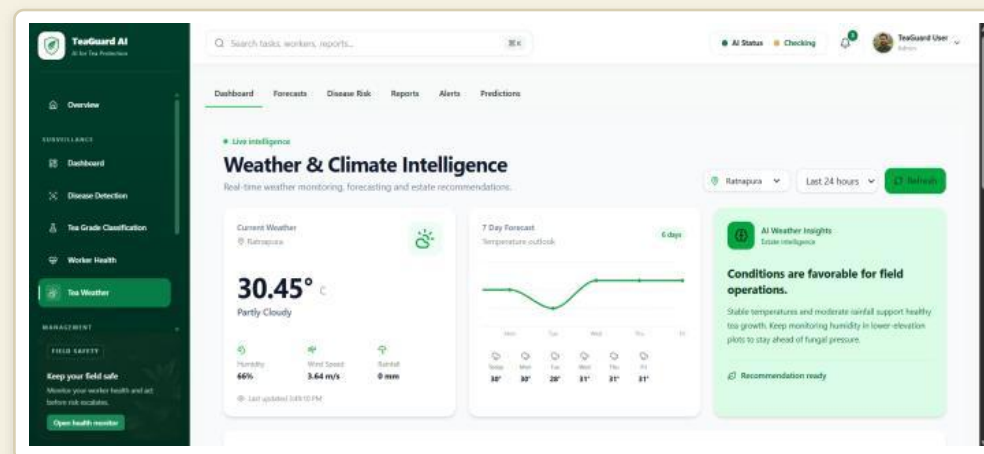


Tea-Specific Weather & Climate Intelligence

Turns raw weather readings into forward-looking guidance — forecasting temperature and tea-disease risk so managers can act before damage becomes visible.

SOLUTION OVERVIEW & ARCHITECTURE

- Covers three tea-growing regions: Nuwara Eliya, Kandy, Ratnapura
- Random Forest predicts disease risk from humidity + temperature thresholds
- Linear/Gradient Boosting models forecast temperature trends
- 8-hour rolling forecast with region-specific recommendations



Live dashboard preview

89%

Prediction Accuracy

Gradient Boosting + Random Forest

Dataset: Historical & real-time regional weather data

Weather forecast · Disease risk · Irrigation & harvest advice

04

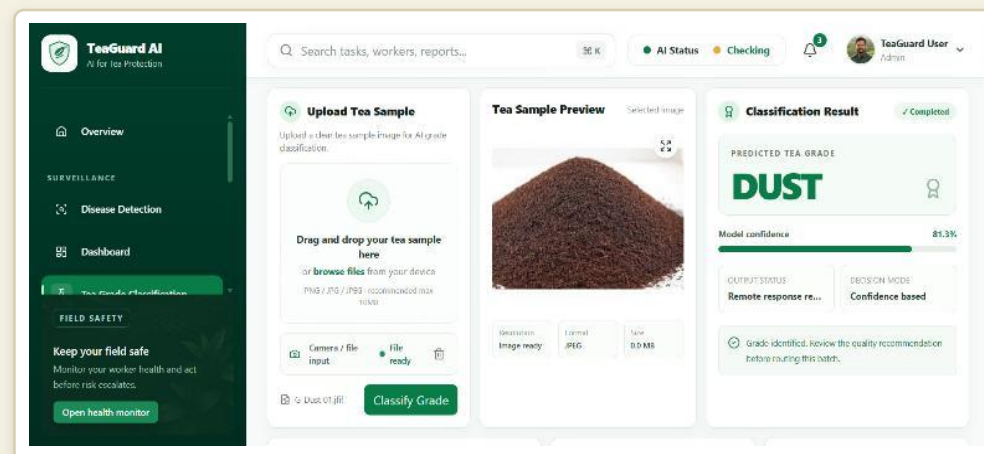


AI-Based Tea Sorting & Grading Automation

Automates the grading of processed tea, replacing subjective manual inspection with consistent, deep-learning-based classification for faster, more reliable quality control.

SOLUTION OVERVIEW & ARCHITECTURE

- User uploads an image of the processed tea sample
- ConvNeXtV2-Tiny deep learning model classifies tea grade
- Grade categories include BM, BOP, BP, DUST, FANNING and more
- Prediction + confidence score stored and shown on dashboard



Live dashboard preview

99.7%

Avg. F1-Score

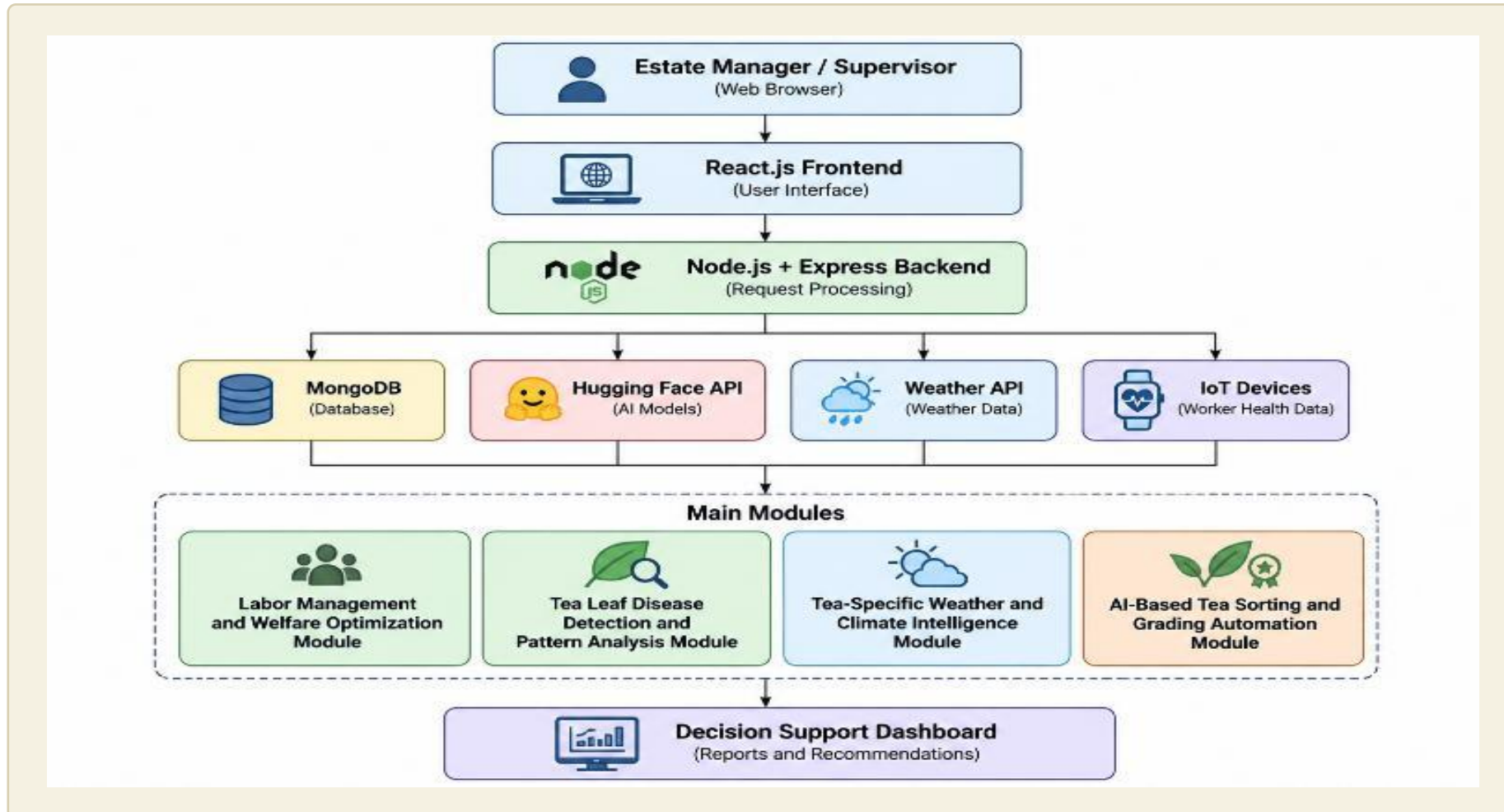
ConvNeXtV2-Tiny

Dataset: Processed Tea Grade Image Dataset

99.7% Avg. Precision • 99.7% Avg. Recall

How the Four Modules Connect

Estate managers use one React.js dashboard; a Node.js/Express backend routes requests to MongoDB, the Hugging Face model API, the weather API, and connected IoT devices.



Where We Stand Today

Labor Management & Welfare Module

Developed & tested — 82% accuracy achieved



Tea Leaf Disease Detection Module

Developed & validated — 96.1% mAP@0.5



Weather & Climate Intelligence Module

Models trained — dashboard integration in progress



Tea Sorting & Grading Module

Developed & validated — 99.7% F1-score



Dashboard Integration (all 4 modules)

Core flows connected — refining alerts & reports



Field Validation with Wewawatte Estate

Pilot testing underway with real estate data



Validated Results Across All Modules

82%

Worker stress classification
accuracy

LightGBM · PhysioNet dataset

96.1%

Disease detection mAP@0.5

YOLOv8s · 95.2% precision, 93.8%
recall

89%

Weather & disease-risk
prediction accuracy

Gradient Boosting + Random Forest

99.7%

Tea grading average F1-score

ConvNeXtV2-Tiny · grade classes

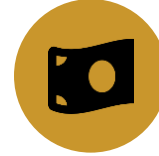
Every module was evaluated independently on its own dataset before dashboard integration each meeting or exceeding comparable literature benchmarks.

Turning TeaGuard AI Into a Product



Target Market

Small-to-large tea estates across Nuwara Eliya, Kandy and Ratnapura, plus estate management companies seeking centralized digital oversight.



Revenue Model

Tiered SaaS subscription per estate (dashboard + AI modules) bundled with optional IoT wearable hardware sales and support.



Go-to-Market

Pilot and refine with Wewawatte Estates Co. and the Tea Research Institute (TRI) before wider estate-network rollout.



Scalability

Modular architecture allows extension to other crops and additional regions with minimal retraining of existing models.

Proving It Works in the Field



Technical Validation

- Each module trained & tested independently on its own dataset
- Standard metrics used: accuracy, precision, recall, F1-score, mAP
- Results benchmarked against comparable literature studies
- Dashboard-level integration testing across all four modules

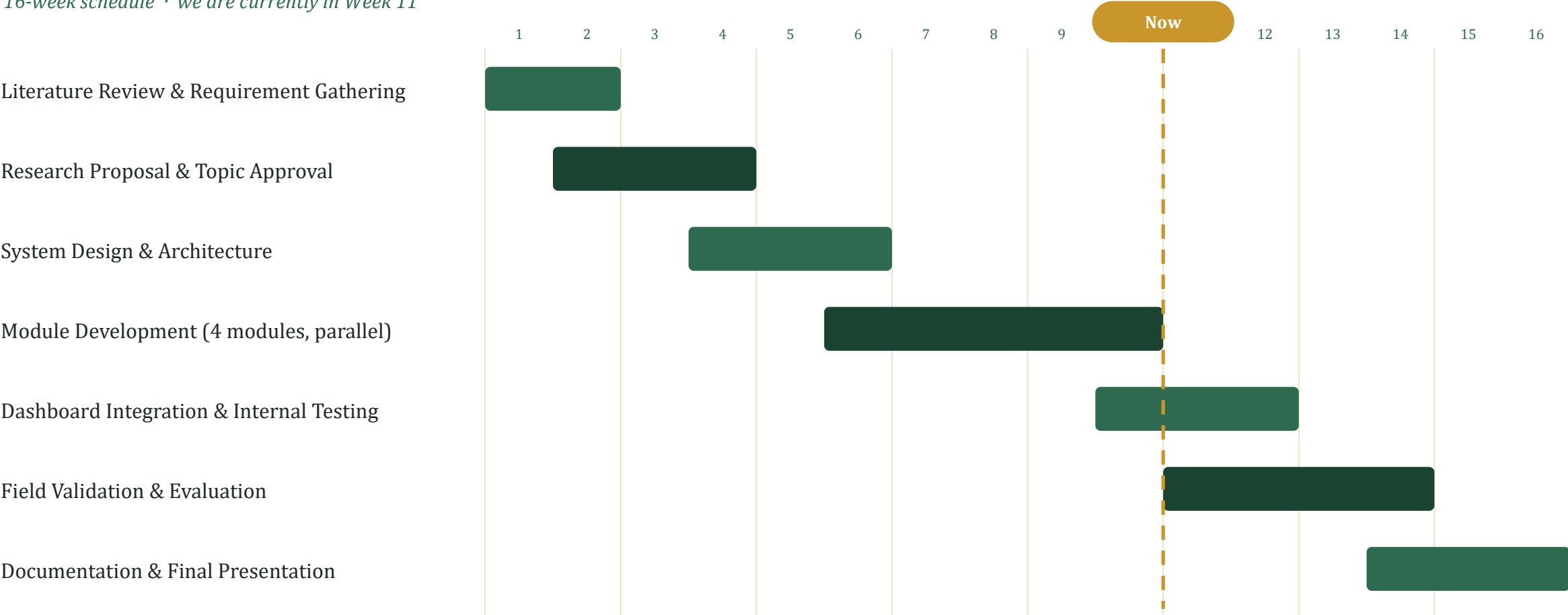


Stakeholder Validation

- Guided by the Tea Research Institute of Sri Lanka (TRI) on disease and climate accuracy
- Real operational data and field feedback provided by Wewawatte Estates Co. (Pvt) Ltd.
- Ongoing pilot deployment to confirm usability for managers, supervisors and field officers
- Findings feeding directly into the next development sprint

Project Timeline

16-week schedule · we are currently in Week 11



09 — RESEARCH SUBMISSIONS

Submissions

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Sri Lanka's tea industry is a major contributor to the national economy and is recognized worldwide for producing high-quality Ceylon tea. However, tea estate management faces challenges such as labour shortages, tea leaf diseases, changing weather conditions, and manual decision-making. This research proposes an Intelligent Decision Support System that integrates Artificial Intelligence, Machine Learning, Deep Learning, and IoT technologies to improve productivity, worker welfare, tea quality, and overall estate management.



Thank You

Questions & Discussion